

Introduction to Kernel Density Estimation for Multivariate Data

DE for Multivariate Data

For a d -variate random sample $X_1, X_2, \dots, X_n$ drawn from a density $f()$, the kernel density estimate is defined by

$$\hat{f}(x, H) = \frac{1}{n} \sum_{i=1}^n K_H(x - X_i)$$

where $K_H(x) = |H|^{-1/2} K(H^{-1/2}x)$, $x = (x_1, x_2, \dots, x_d)$, and $X_i = (X_{i1}, X_{i2}, \dots, X_{id})$, $i = 1, 2, \dots, n$. Here $K(x)$ is the kernel which is a symmetric probability density function, H is the bandwidth matrix which is symmetric and positive-definite.

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where $K_H(x) = |H|^{-1/2} K(H^{-1/2}x)$.

- the choice of K is not crucial: $k(x) = (2\pi)^{-d/2} \exp(-\frac{1}{2}x^T x)$
- the choice of H is crucial

Example

A mixture density:

$$\frac{4}{11}N(\mu_1, I) + \frac{3}{4}N(0, \Sigma_2) + \frac{4}{11}N(\mu_3, I)$$

where $\mu_1 = [-2, 2]^T$, $\mu_3 = [2, -2]^T$, and

$$\Sigma_2 = \begin{bmatrix} 0.8 & -0.72 \\ -0.72 & 0.8 \end{bmatrix}$$

Example

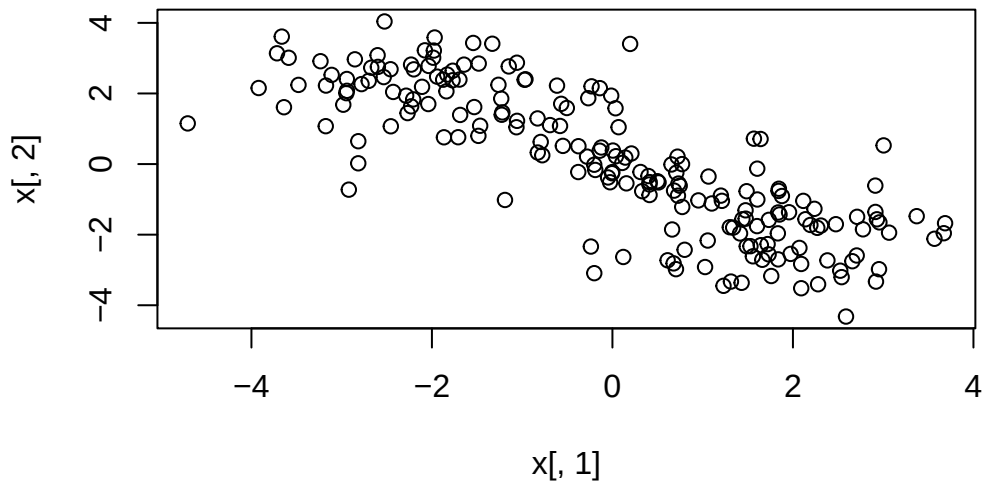
A mixture density:

$$\frac{4}{11}N(\mu_1, I) + \frac{3}{4}N(0, \Sigma_2) + \frac{4}{11}N(\mu_3, I)$$

```
library(mvtnorm)
library(ks)
```

Warning: package 'ks' was built under R version 4.3.3

```
samp <- 200
mus <- rbind(c(-2,2), c(0,0), c(2,-2))
Sigmas <- rbind(diag(2), matrix(c(0.8, -0.72, -0.72, 0.8), nrow = 2), diag(2))
cwt <- 3/11
props <- c((1-cwt)/2, cwt, (1-cwt)/2)
x <- rmvnorm.mixt(n = samp, mu = mus, Sigma = Sigmas, props = props)
plot(x[,1], x[,2])
```



bandwidth matrix

- diagonal: $diag(h_1^2, \dots, h_d^2)$
- unconstrained: a symmetric and positive-definite matrix

```
library(ks)
library(plot3D)
```

Warning: package 'plot3D' was built under R version 4.3.3

```
H_diag <- Hpi.diag(x = x)
H_uncond <- Hpi(x = x)
newx_diag <- kde(x = x, H = H_diag)
newx_uncond <- kde(x = x, H = H_uncond)
#persp(x=newx_diag$eval.points[[1]],
#      y=newx_diag$eval.points[[2]],
#      z=newx_diag$estimate)
persp3D(x=newx_diag$eval.points[[1]], main="diagonal",
        y=newx_diag$eval.points[[2]],
        z=newx_diag$estimate, theta = 2, colkey = F)
persp3D(x=newx_uncond$eval.points[[1]],
        y=newx_uncond$eval.points[[2]], main="unconstrained",
        z=newx_uncond$estimate, theta = 0, colkey = F)
```

Pre-transformation

Instead of basing our bandwidth selection on the original data $X_1, X_2, \dots, X_n$, we could use the transformed data $X_1^*, X_2^*, \dots, X_n^*$, where the transformation is either sphering

$$X^* = S^{-1/2} X$$

where S is the sample covariance matrix of the untransformed data; or scaling

$$X^* = S_D^{-1/2} X$$

where $S_D = \text{diag}(s_1^2, \dots, s_d^2)$.

Pre-transformation

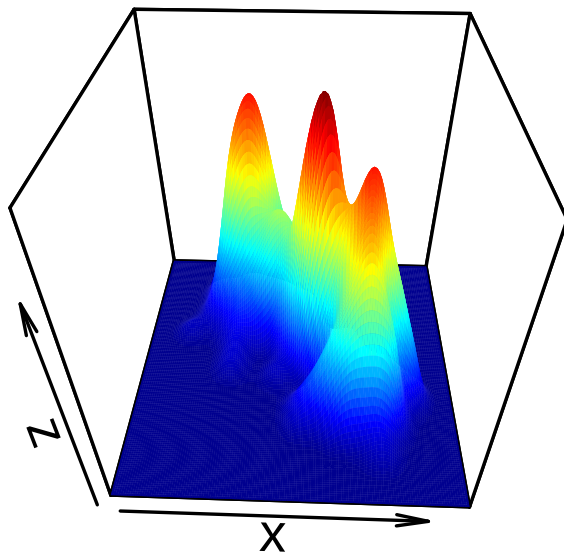
The bandwidth matrix H^* suitable for the sphered or scaled data can be back transformed to the original scale by

$$H = S^{1/2} H^* S^{1/2}$$

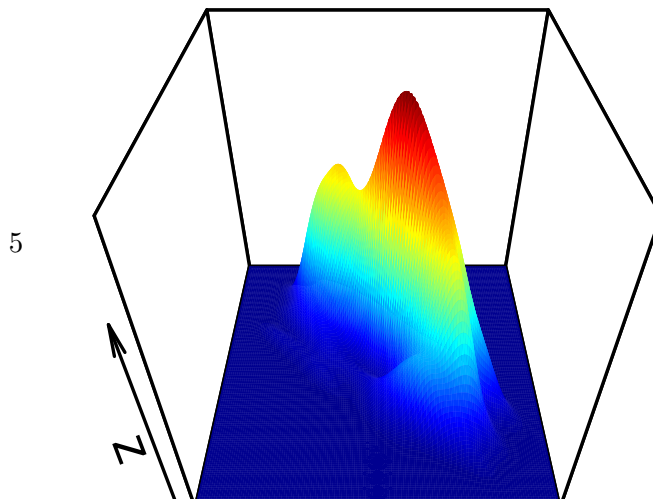
or

$$H = S_D^{1/2} H^* S_D^{1/2}$$

diagonal



unconstrained



Optimal bandwidth matrix

the measure the performance (Mean Integrated Squared Error, MISE):

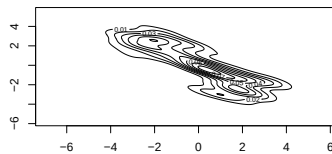
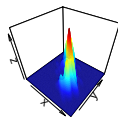
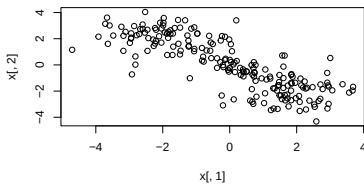
$$MISE(H) = E \int [\hat{f}(x; H) - f(x)]^2 dx$$

- Plug-in
- Cross validation (LSCV)
- Smoothed cross validation (SCV)

Example

$$\frac{4}{11}N(\mu_1, I) + \frac{3}{4}N(0, \Sigma_2) + \frac{4}{11}N(\mu_3, I)$$

```
Hlscv1 <- Hlscv(x = x)
newx1 <- kde(x = x, H = Hlscv1)
x.new <- newx1$eval.points[[1]]
y.new <- newx1$eval.points[[2]]
z <- newx1$estimate
plot(x[,1], x[,2])
persp3D(x.new, y.new, z, colkey = F)
contour(x=x.new, y=y.new, z=z)
```



Kernel discriminant analysis

We assume that X has a categorical random variable Y as a companion, the labels of Y indicating one out of m possible classes within the population X , and that we have a sample $(X_1, Y_1), \dots, (X_n, Y_n)$.

KD Classification

Let's denote f_j to the conditional pdf of $X|Y = j$ and π_j to the value of the probability mass function of Y at j , i.e., $P\{Y = j\} = \pi_j$, for $j = 1, \dots, m$. In this framework, the unconditional pdf f of X is a mixture of the conditional pdfs.

$$f(x) = \sum_{j=1}^m f_j(x) \pi_j$$

Classification

The problem of assigning an observation x of X to each of the classes of Y can be tackled by maximizing the conditional probability

$$P\{Y = j | \mathbf{X} = \mathbf{x}\} = \frac{f_j(\mathbf{x}) P\{Y = j\}}{\sum_{j=1}^m f_j(\mathbf{x}) P\{Y = j\}} \quad (0.1)$$

$$= \frac{f_j(\mathbf{x}) \pi_j}{\sum_{j=1}^m f_j(\mathbf{x}) \pi_j} \quad (0.2)$$

Bayes classifier

The Bayes classifier exploits this idea of assigning $\mathbf{x}$ to the most likely class j .

$$\arg \max_{j=1, \dots, m} \pi_j f_j(\mathbf{x})$$

- Estimating the class probabilities π_j is simple: $\hat{p}_j = n_j/n$
- Estimating f_j can be done by a kernel density estimation

kernel discriminant analysis

We assume that X has a categorical random variable Y as a companion, the labels of Y indicating one out of m possible classes within the population X , and that we have a sample $(X_1, Y_1), \dots, (X_n, Y_n)$.

$$\gamma(\mathbf{x}; H_1, \dots, H_m) = \arg \max_{j=1, \dots, m} \hat{\pi}_j \hat{f}_j(\mathbf{x}; H_j)$$

The kernel classifier needs to be fed by the bandwidths $H_1, \dots, H_m$, each of them to be selected from each of the m subsamples.

Bivariate Example: iris

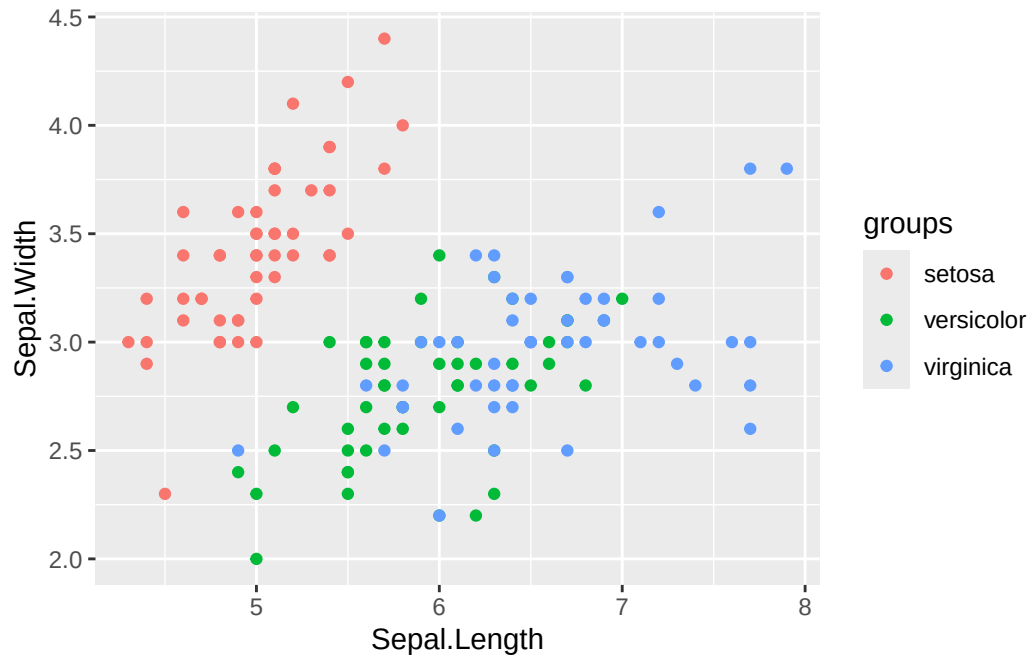
```
x <- iris[, 1:2]
head(x, 3)
```

	Sepal.Length	Sepal.Width
1	5.1	3.5
2	4.9	3.0
3	4.7	3.2

```
groups <- iris$Species
prop.table(table(groups))
```

```
groups
      setosa versicolor  virginica
0.3333333  0.3333333  0.3333333
```

```
library(ggplot2)
dt <- data.frame(x, groups = as.factor(groups))
ggplot(iris, aes(x=Sepal.Length,
                 y=Sepal.Width, color=groups))+
  geom_point()
```

Bivariate Example: iris

```
library(ks)
#By default, the ks::Hpi bandwidths are computed
kda_pi <- ks::kda(x = x, x.group = groups)
table(kda_pi$x.group.estimate,
      groups, dnn=c("Est","True"))
```

	True		
Est	setosa	versicolor	virginica
setosa	50	0	0
versicolor	0	37	11
virginica	0	13	39

Bivariate Example: iris

```
#By default, the ks::Hpi bandwidths are computed
Hs <- ks::Hkda(x=x, x.group = groups, bw="scv")
kda_scv <- ks::kda(x = x, x.group = groups, Hs = Hs)
table(kda_scv$x.group.estimate,
      groups, dnn=c("Est","True"))
```

	True		
Est	setosa	versicolor	virginica
setosa	50	0	0
versicolor	0	37	11
virginica	0	13	39

Bivariate Example: iris

- Classification of new observations

```
newx <- x[c(5,55,105), ]
newx
```

	Sepal.Length	Sepal.Width
5	5.0	3.6
55	6.5	2.8
105	6.5	3.0

```
predict(kda_scv, x = newx)
```

```
[1] setosa    virginica virginica
Levels: setosa versicolor virginica
```

```
groups[c(5,55,105)]
```

```
[1] setosa    versicolor virginica
Levels: setosa versicolor virginica
```

Classification error

```
table(groups,
      kda_scv$x.group.estimate,
      dnn=c("True", "Est"))
```

	Est		
True	setosa	versicolor	virginica
setosa	50	0	0
versicolor	0	37	13
virginica	0	11	39

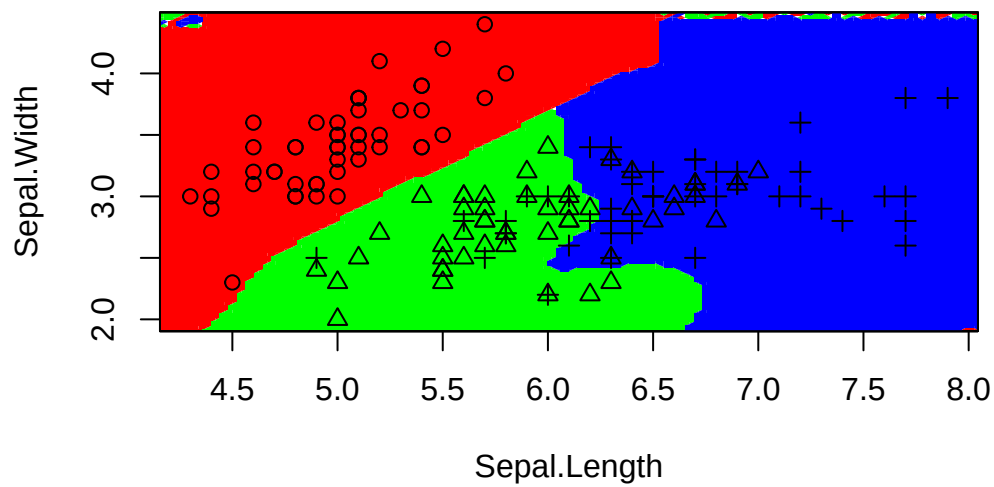
```
ks::compare(x.group = kda_scv$x.group,
            est.group = kda_scv$x.group.estimate)
```

	setosa (est.)	versicolor (est.)	virginica (est.)	Total
setosa (true)	50	0	0	50
versicolor (true)	0	37	13	50
virginica (true)	0	11	39	50
Total	50	48	52	150

```
$error
[1] 0.16
```

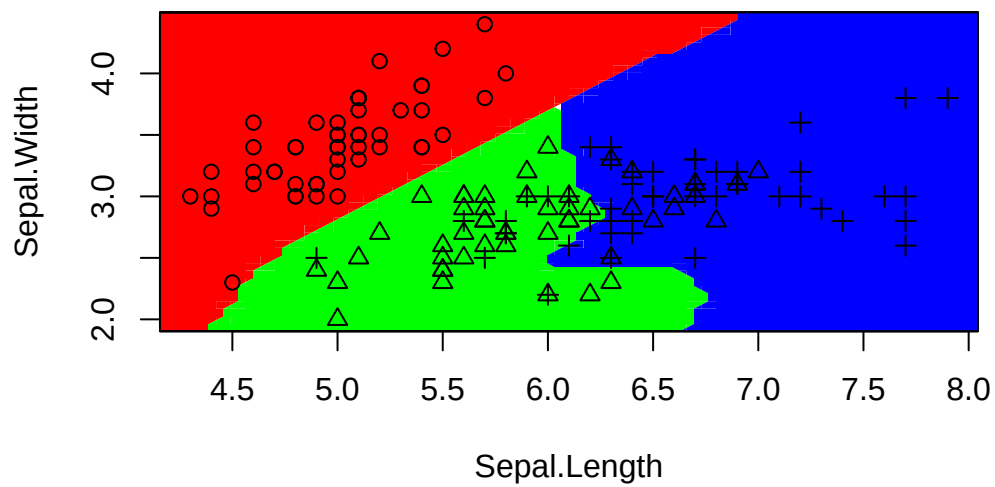
Plot of classification regions

```
plot(kda_scv, col = rainbow(3),
     lwd = 2, col.pt = 1, cont=FALSE,
     col.part = rainbow(3, alpha = 0.25), drawpoints = TRUE)
```



Plot of classification regions

```
# supp(3.7): effective support for standard normal
kda_3 <- ks::kda(x = x, x.group = groups, Hs = Hs, supp = 10)
plot(kda_3, col = rainbow(3),
     col.pt=1, cont=FALSE,
     col.part = rainbow(3, alpha = 0.25), drawpoints = TRUE)
```



LDA and QDA

```
lda <- MASS::lda(Species ~ ., data=iris[,c(1,2,5)])  
qda <- MASS::qda(Species ~ ., data=iris[,c(1,2,5)])  
boundary(lda, iris[,c(1,2,5)], class = "Species")  
boundary(qda, iris[,c(1,2,5)], class = "Species")
```

